

IMU Sensor Data Cleaning, Exploration, and Feature Engineering

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1 Introduction

1.1 Dataset Description

You are provided with `imu_messy_data.csv`, a synthetic accelerometer dataset containing recordings from three subjects performing three activities:

1. **still**: Subject is stationary
2. **walk**: Subject is walking
3. **shake**: Subject is shaking the device

The dataset has the following columns:

- **timestamp**: Unix timestamp (seconds since epoch)
- **subject**: Subject identifier (subject_1, subject_2, subject_3)
- **acc_x**: Acceleration in X axis (m/s^2)
- **acc_y**: Acceleration in Y axis (m/s^2)
- **acc_z**: Acceleration in Z axis (m/s^2)
- **label**: Activity label

1.2 Data Quality Issues

The dataset intentionally contains multiple types of real-world problems:

- **Missing values**: Approximately 5% of accelerometer readings are missing (NaN)
- **Outliers and spikes**: About 2% of samples have extreme values (sensor glitches)
- **Duplicates**: Roughly 3% duplicate rows exist
- **Timestamp issues**: 20% of timestamps have jitter/drift, plus one major gap
- **Gaussian noise**: 30% of samples have added noise
- **Label noise**: Approximately 2% of labels are incorrect
- **Unsorted data**: Rows are shuffled (timestamps not in order)
- **Subject variability**: Different subjects have different signal amplitudes

2 Part 1: Data Loading and Initial Exploration

2.1 Task 1.1: Load and Inspect the Data

Requirements:

1. Load `imu_messy_data.csv` into a pandas DataFrame
2. Display the first 20 rows
3. Print dataset shape, column names, and data types
4. Generate summary statistics using `describe()`

Questions to answer:

- a) How many samples are in the dataset?
- b) What is the range of timestamp values?
- c) What are the min/max values for each acceleration axis? Do they seem physically plausible?
- d) What is the class distribution across activities?

2.2 Task 1.2: Missing Value Analysis

Requirements:

1. Count and report missing values per column
2. Calculate the percentage of missing data for each accelerometer axis
3. Identify rows where all three axes are missing simultaneously
4. Visualize the distribution of missing values across the dataset (e.g., missing value heatmap or pattern plot)

Questions to answer:

- a) Which axis has the most missing values?
- b) Are missing values randomly distributed or clustered?
- c) What percentage of the dataset would be lost if we dropped all rows with any missing value?

2.3 Task 1.3: Timestamp Analysis

Requirements:

1. Check if timestamps are sorted in ascending order
2. Calculate the time differences between consecutive samples
3. Identify gaps larger than expected (assuming 50 Hz sampling rate)
4. Detect timestamp duplicates (same timestamp appearing multiple times)

Questions to answer:

- a) What is the expected time interval between samples at 50 Hz?
- b) How many samples have timestamp gaps exceeding 1 second?
- c) What is the largest gap in the data? Where does it occur?
- d) Are there any duplicate timestamps?

3 Part 2: Data Cleaning

3.1 Task 2.1: Handling Missing Values

Implement and compare three strategies for handling missing values:

Strategy A: Forward Fill

- Use `fillna(method='ffill')` to propagate last valid value forward
- Report how many values were filled

Strategy B: Linear Interpolation

- Use `interpolate(method='linear')` for time-aware interpolation
- Compare interpolated vs. original values on a small sample

Strategy C: Drop Rows

- Remove all rows containing any NaN values
- Calculate percentage of data lost

Questions to answer:

- a) Which strategy preserves the most data?
- b) For a segment with missing values, plot the original signal and both filled versions (forward fill and interpolation). Which looks more realistic?
- c) What are the trade-offs of each approach?
- d) Which strategy would you recommend for this dataset? Justify your answer.

3.2 Task 2.2: Outlier Detection and Treatment

Requirements:

1. Implement z-score based outlier detection (threshold: $|z| > 3$)
2. Implement IQR-based outlier detection (values beyond $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$)
3. Count outliers detected by each method for each axis
4. Visualize outliers using box plots and scatter plots over time

Treatment strategies to compare:

1. **Removal:** Delete rows containing outliers
2. **Clipping:** Cap outliers at threshold values
3. **Interpolation:** Replace outliers with interpolated values

Questions to answer:

- a) How many outliers are detected by z-score vs. IQR method?
- b) Which method is more conservative (flags fewer outliers)?
- c) Examine 5 specific outlier samples. Are they legitimate extreme values or clear errors?
- d) For each treatment strategy, calculate the mean and standard deviation of each axis. How do they change?
- e) Which outlier treatment strategy would you choose? Why?

3.3 Task 2.3: Duplicate Removal

Requirements:

1. Identify exact duplicate rows (all columns identical)
2. Identify duplicate measurement rows (timestamp, acc_x, acc_y, acc_z identical)
3. Remove duplicates keeping the first occurrence
4. Document how many duplicates were found and removed

Questions to answer:

- a) How many exact duplicates exist?
- b) How many measurement duplicates (ignoring subject/label) exist?
- c) Should duplicate removal happen before or after sorting by timestamp? Why?

3.4 Task 2.4: Timestamp Sorting and Resampling

Requirements:

1. Sort the dataset by timestamp in ascending order
2. Set timestamp as the DataFrame index (convert to datetime if needed)
3. Resample to a fixed 50 Hz rate using appropriate interpolation
4. Handle the major gap in the data appropriately

Questions to answer:

- a) After sorting, visualize a 5-second window of each axis. Does the signal appear continuous?
- b) How should the major gap be handled? (options: keep as is, interpolate, split into separate sessions)
- c) After resampling, verify the sampling rate. Is it exactly 50 Hz?

4 Part 3: Exploratory Data Analysis and Visualization

4.1 Task 3.1: Time-Series Visualization

Requirements:

1. Plot a 10-second segment of all three axes for each activity class
2. Create a multi-panel figure showing one representative segment per activity
3. Color-code by activity label and add appropriate legends
4. Plot the acceleration magnitude: $||a|| = \sqrt{a_x^2 + a_y^2 + a_z^2}$

Questions to answer:

- a) What are the characteristic patterns for each activity?
- b) Which activity has the highest variance in acceleration?
- c) Can you visually identify activity transitions?
- d) Does the acceleration magnitude provide better class separation than individual axes?

4.2 Task 3.2: Statistical Analysis per Class

Requirements:

1. Calculate mean, standard deviation, min, and max for each axis grouped by activity
2. Create box plots for each axis grouped by activity
3. Plot histograms of each axis for each activity (3 axes $\times$ 3 activities = 9 plots)
4. Calculate correlation matrices for each activity

Questions to answer:

- a) Which activity has the highest mean acceleration in the Z axis? Why?
- b) Which features (mean, std, etc.) show the most separation between activities?
- c) Are the three axes correlated within each activity?
- d) Based on these statistics, which activities should be easiest/hardest to classify?

4.3 Task 3.3: Subject Variability Analysis

Requirements:

1. Calculate mean acceleration magnitude for each subject and activity combination
2. Create a grouped bar chart showing subject differences
3. Compute coefficient of variation ($CV = \text{std}/\text{mean}$) for each subject

Questions to answer:

- a) Do different subjects show significantly different signal amplitudes?
- b) Should we normalize per-subject before training a classifier? Why or why not?
- c) How would you split this data for train/test? Random split or by subject?

4.4 Task 3.4: Label Noise Investigation

Requirements:

1. For each activity, calculate the mean and std of acceleration magnitude
2. Identify samples whose magnitude deviates significantly from their class mean
3. Manually inspect 10-20 suspicious samples (plot their neighborhood in time)

Questions to answer:

- a) Can you identify likely mislabeled samples based on signal characteristics?
- b) What percentage of labels appear to be noisy?
- c) How would incorrect labels affect a machine learning model?

5 Part 4: Preprocessing and Segmentation

5.1 Task 4.1: Normalization Strategies

Implement and compare three normalization approaches:

Approach 1: Per-Axis Standardization (Z-score)

- Normalize each axis to zero mean and unit variance: $x' = (x - \mu)/\sigma$

Approach 2: Min-Max Scaling

- Scale each axis to range $[0, 1]$: $x' = (x - x_{min})/(x_{max} - x_{min})$

Approach 3: Per-Subject Standardization

- Standardize each axis separately for each subject

Questions to answer:

- a) Plot the same 5-second window before and after each normalization. How do they differ?
- b) Which approach preserves relative differences between activities best?
- c) Why might per-subject normalization be important?
- d) What are the risks of min-max scaling with outliers still present?

5.2 Task 4.2: Sliding Window Segmentation

Requirements:

1. Implement sliding window segmentation with parameters:
 - Window size: 2 seconds (100 samples at 50 Hz)
 - Overlap: 50% (50 samples)
2. Assign label to each window (majority voting if labels differ within window)
3. Count total number of windows per activity
4. Experiment with different window sizes: 1s, 2s, 4s

Questions to answer:

- a) How many windows are generated with 2s window and 50% overlap?
- b) What happens at activity transitions? How would you handle windows spanning two activities?
- c) Compare window sizes: what are the trade-offs of shorter vs. longer windows?
- d) What overlap percentage would you recommend? Why?

5.3 Task 4.3: Handling Gaps and Missing Segments

Requirements:

1. Identify the major gap in the data
2. Split the dataset into continuous sessions (before gap and after gap)
3. Report the duration of each session
4. Decide on a strategy: treat as separate sessions or interpolate

Questions to answer:

- a) How long is the gap? Is interpolation reasonable?
- b) Should windows spanning the gap be discarded?
- c) How would this gap affect model training and evaluation?

6 Part 5: Feature Engineering

6.1 Task 5.1: Time-Domain Features

For each sliding window, extract the following features for each axis (acc_x, acc_y, acc_z):

Basic statistics:

- Mean: $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$
- Standard deviation: $\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2}$
- Min and Max values
- Range: $x_{max} - x_{min}$

Signal characteristics:

- Signal magnitude area (SMA): $\frac{1}{N} \sum_{i=1}^N |x_i|$
- Energy: $\frac{1}{N} \sum_{i=1}^N x_i^2$
- Zero-crossing rate (ZCR): Count of times signal crosses zero
- Root mean square (RMS): $\sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$

Magnitude features:

- Mean acceleration magnitude: $\bar{m} = \frac{1}{N} \sum_{i=1}^N \sqrt{x_i^2 + y_i^2 + z_i^2}$
- Std of magnitude

Requirements:

1. Create a feature matrix where each row is one window
2. Document the total number of features extracted
3. Create a DataFrame with features and corresponding labels

Questions to answer:

- a) How many total features do you have per window?
- b) Which time-domain features show the best separation between classes? (use box plots)
- c) Are some features redundant (highly correlated)?

6.2 Task 5.2: Frequency-Domain Features

For each sliding window, apply FFT and extract:

FFT features:

- Dominant frequency (frequency with highest magnitude)
- Spectral energy (sum of squared FFT magnitudes)
- Spectral entropy
- Power in frequency bands (0-2 Hz, 2-5 Hz, 5-10 Hz, 10-25 Hz)

Requirements:

1. Apply FFT to each axis in each window
2. Compute frequency features for each axis
3. Visualize power spectral density for representative windows of each activity

Questions to answer:

- a) What is the dominant frequency for each activity?
- b) Which frequency band contains the most energy for walking? For shaking?
- c) Do frequency features provide better class separation than time-domain features?
- d) How do window size and overlap affect frequency resolution?

6.3 Task 5.3: Feature Selection and Analysis

Requirements:

1. Calculate correlation matrix for all features
2. Identify highly correlated feature pairs (—correlation— ≥ 0.9)
3. Compute feature importance (e.g., using mutual information or ANOVA F-score)
4. Select top 10 most informative features

Questions to answer:

- a) How many feature pairs are highly correlated?
- b) Which features are most informative for classification?
- c) If you had to choose only 5 features, which would you pick? Why?
- d) Would you use PCA for dimensionality reduction here? Why or why not?

7 Tips

7.1 Python Libraries

You will need:

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 from scipy import signal, stats
6 from scipy.fft import fft, fftfreq
7 from sklearn.preprocessing import StandardScaler, MinMaxScaler
```